

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Evaluate the limit and justify any algebraic simplification.

[4 marks]

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

2

Differentiate the function and simplify your answer.

[5 marks]

$$y = (3x^2 + 1)^5$$

3

A rectangle has perimeter 40 cm. Find the dimensions that maximise its area, and show your result is a maximum.

[6 marks]

4

Evaluate the definite integral, showing the antiderivative you use.

[5 marks]

$$\int_0^2 (3x^2 - 4x + 1) dx$$

5

Find the equation of the tangent to the curve below at the point where $x = 2$.

[5 marks]

$$y = x^3 - 2x$$

6

Solve the differential equation given that $y(0)=2$.

[6 marks]

$$\frac{dy}{dx} = 3y$$

7

Decompose into partial fractions, stating the form you assume before finding the coefficients.

[6 marks]

$$\frac{5x + 1}{(x - 1)(x + 2)}$$

8

Integrate by parts, stating your choice of u and dv .

[5 marks]

$$\int x e^x dx$$
